GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 5

COURSE: COMPUTER NETWORKS (ETCS-503)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETCS-503

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Differentiate between OSI 7-layer reference model and TCP/IP protocol suite.

Q1. (b) Calculate channel capacity using Nyquist formula and Shannon-Hartley theorem.

Q1. (c) What is bit stuffing and byte stuffing in data link framing?

Q1. (d) What is Hamming code? Calculate minimum Hamming distance to detect d errors.

Q1. (e) Explain Pure ALOHA vs Slotted ALOHA with maximum throughput values.

Q1. (f) Differentiate between IPv4 and IPv6 header formats.

Q1. (g) What is subnet mask? Given IP 192.168.1.0/26, find total subnets and usable hosts per subnet.

Q1. (h) Why does TCP require Three-Way Handshake for connection establishment?

Q1. (i) What is difference between Distance Vector Routing and Link State Routing?

Q1. (j) Explain DNS resolution process (Iterative vs Recursive).

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Explain Sliding Window Protocol: Compare Stop-and-Wait, Go-Back-N, and Selective Repeat with window sizes.

Q2. (b) Perform CRC error detection for dataword 1101001110 using divisor polynomial G(x) = x^4 + x + 1.

Q2. (c) A company is assigned network 200.10.10.0/24. Subnet it into 4 departments requiring 50, 25, 25, 10 hosts using VLSM.

Q2. (d) Explain TCP Congestion Control mechanisms: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery.

Q2. (e) Explain Bellman-Ford Distance Vector routing algorithm and explain Count-to-Infinity problem and solutions.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain CSMA/CD protocol with binary exponential backoff algorithm in Ethernet. [10 Marks]

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(b) Explain CSMA/CA protocol with RTS/CTS virtual carrier sensing in IEEE 802.11 Wireless LAN. [10 Marks]

Q4. (a) Explain IPv4 datagram header fields: TTL, Identification, Flags, Fragment Offset, and Checksum. [10 Marks]

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(b) Explain Address Resolution Protocol (ARP) and Reverse ARP (RARP) with packet formats. [10 Marks]

Q5. (a) Explain OSPF (Open Shortest Path First) protocol: Link state advertisements, Dijkstra shortest path, and areas. [10 Ma

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(b) Explain BGP (Border Gateway Protocol) path vector routing in autonomous systems. [10 Marks]

Q6. (a) Explain TCP flow control using sliding window with Silly Window Syndrome and Nagle algorithm. [10 Marks]

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(b) Compare TCP and UDP based on connection orientation, header size, reliability, and use cases. [10 Marks]

Q7. (a) Explain HTTP/1.1 vs HTTP/2 vs HTTP/3 (QUIC) protocols with multiplexing and head-of-line blocking. [10 Marks]

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(b) Explain RSA public key encryption algorithm with a numerical example using p=7, q=11, e=13. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****